Ajoy Dey

LinkedIn | GitHub | Email: ajoydey00001@gmail.com | Mobile: +8801827390358

Research Interests: Computer System, Cybersecurity, Machine Learning

Education

Bangladesh University of Engineering and Technology — Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering; CGPA 3.89/4.0 — *Feb. 2020 – March 2025*

Chattogram College — Chittagong, Bangladesh
HSC; GPA 5.0/5.0 — *2019*

Uttar Kattali Al-Haj Mostafa Hakim High School — Chittagong, Bangladesh
SSC; GPA 5.0/5.0 — *2017*

Technical Skills

Language: C, C++, Bash, LATEX, SQL, Assembly ,Golang, Java, Python, JavaFX
Web Technologies: HTML, CSS, JavaScript, React.js, Node.js
Library: Sklearn, Pandas, Matplotlib, Seaborn
Database Systems: Oracle, MongoDB
Other Tools: IntelliJ IDEA, Visual Studio Code, Docker, Wireshark

Computer System Projects

Compiler Offlines — Github
CSE 310 Compiler Construction Assignments

- Offline 01: Implemented a scoped symbol table in C++ using hash tables and nested scope management.
- Offline 02: Built a Flex lexical analyzer for a C-like language with tokenization and error reporting.
- Offline 03: Developed a YACC/Bison parser with semantic analysis, function handling, and parse-tree generation.
- Offline 04: Added intermediate code generation and simple assembly optimization for the compiler pipeline.

Operating Systems Sessional Offlines — GitHub
CSE 314

- Developed a shell script to organize, compile, execute, and evaluate student submissions across C, Java, and Python.
- Implemented custom xv6 system calls for syscall tracing, usage history, and controlled QEMU termination.
- Modified the xv6 scheduler with a two-level MLFQ using lottery scheduling, round-robin scheduling, and priority boosting.
- Built a POSIX thread-based IPC simulation using semaphores and reader-writer synchronization.

Computer Security Offlines
Github

- Project 01: Implemented a cryptosystem that uses a symmetric key for encryption and decryption. The symmetric key is securely shared by the Elliptic Curve Diffie Hellman key exchange method.
- Project 02: Cross-Site Scripting (XSS) Attack
- Project 03: Buffer Overflow Attack

Football Player Database System — Github
It is built on Java, JavaFX with a server-client architecture, allowing clients to view and buy players.

- The server reads player data from a file, handles client connections, and updates player lists based on client interactions.

Software Projects

Food Bear

This website built on the MERN stack which allows users to order food from various restaurants conveniently. *Github*

- Developed a food delivery website supporting restaurant search, order tracking, cart management, and home kitchen functionality.
- Integrated real-time features like order tracking, restaurant analytics dashboard, and voucher-based discounts.
- Implemented Google Maps integration for location-based restaurant search.

Thesis

Sentence-level Bangla Spell Error Corrector

My Undergraduate Thesis is supervised by ***Dr. Muhammad Abdullah Adnan sir*** *GitHub*

- Developed a seq2seq transformer model ("MarianMT") fine-tuned on 3.8M sentence pairs generated using the "Synthetic Error Dataset Generation Mimicking Bengali Writing Pattern" algorithm.
- Utilized pre-trained checkpoints from Bangla Grammatical Error Correction (BGEC) tasks for knowledge transfer.
- Achieved 97% character-level accuracy, exploring Knowledge Distillation to improve performance further.

Experience

Software Engineer I – Pathao Bangladesh

Full-time *May 2025 – Present*

- Implemented a new routing algorithm in a backend ride service to provide highly reliable ride routes, improving route reliability to 90% for over 100,000 rides completed daily.
- Developed and deployed an MLOps pipeline for ride ETA (pickup-to-dropoff) prediction using live timestamp datasets, achieving 85% prediction accuracy in production.
- Developed and deployed an Autonomous pipeline with N&N which gives overall result of every day to next day through emails.
- Worked on backend system optimization and large-scale ride data processing to improve service efficiency and scalability.

Achievements

- **Merit List (BUET, 2023)** – Secured a place in the BUET Merit List.
- **Dean's List (BUET, 2021, 2022)** – Recognized for outstanding academic performance.
- **Board General Scholarship (HSC, 2019)**
- **Board General Scholarship (SSC, 2017)**

References

Dr. Muhammad Abdullah Adnan
Professor
Department of CSE, BUET
Email: abdullah.adnan@gmail.com
Phone: 01552336926

Dr. Mohammad Saifur Rahman
Professor
Department of CSE, BUET
Email: mrahman@gmail.com